Embedding Dimension & Representation

- Embedding Model: all-MiniLM-L6-v2
- Vector Dimensionality: 384 (small and quick, also our dataset is also small)
- Handling Long Sequences:
  - Context Limit: 256-token (~ 200 words)
  - Structured Payload: Title: {title} | Artist: {artist} | Lyrics: { around first 200 words} (cutting out repetitive outro loops)

Similarity Metric

- Selected Metric: Cosine Similarity
- Normalization Scheme: L2
- Justification:
  - Dataset is not too large, so the vectors are not too crammed
  - When embedding a query, the different between the length of the query and the payload are so big that if we didn’t normalized, the difference in magnitude could result in bad searches.

Indexing Strategy

- Vector indexing: Brute-force/Flat
- Justify across trade-offs:
  - 100% accurate
    - meaning compare the query vector to each of the songs' vectors
  - Latency
    - Since the dataset is small, brute-forcing through 57k songs is quick (only 57000 x 384 calculations per query, a computer can calculate 100-200B calculations per second)
  - Memory Footprint: ~84.5 MB Total
    - 57,650 × 384 × 4 bytes = 84.5 MB when loading into ram